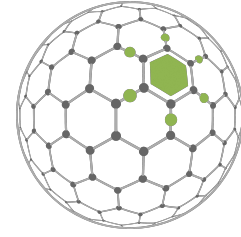


SYSTEMS MANAGEMENT: FROM BOOT-UP TO RUNTIME

Open Atomic Ethernet Proposal for the

Chiplet Summit 2026



The shift toward chiplet-based architectures in modern computing platforms—especially for AI/ML, HPC, and hyperscale datacenters—introduces profound challenges and opportunities in systems management. This essay examines the end-to-end lifecycle of system management from cold boot to runtime, through the lens of heterogeneous chiplet integration, and proposes a new systems-level design philosophy to support emerging reliability, observability, and configurability demands.

INTRODUCTION

Traditional datacenter systems management has relied on a clear layering of firmware, OS, and runtime orchestrators with predictable roles. But the rise of modular, chiplet-based systems—where CPUs, GPUs, NPUs, memory controllers, and interconnect fabrics are disaggregated and recombined—requires rethinking these assumptions.

Chiplet systems blur the boundaries between “platform” and “application”, necessitating tighter coordination between hardware boot-time infrastructure and runtime services. What once happened in firmware alone—e.g., power negotiation, memory training, error detection—may now require collaboration across multiple dies, vendors, and physical packages. Similarly, runtime telemetry, orchestration, and security models must evolve to track dynamic reconfiguration, per-die fault containment, and low-latency control loops operating close to the wire.

BOOT-UP: ESTABLISHING THE INITIAL TRUST CHAIN

The first moment of life in a chiplet system is a distributed event. Unlike monolithic SoCs, which boot in a well-defined order under centralized control, multi-die systems must negotiate power domains, establish communication links, and synchronize multiple firmware agents that may not share a common root of trust.

KEY CHALLENGES AT BOOT TIME

- **Secure Boot Across Chiplets:** Each chiplet may carry its own firmware, from different vendors. Boot-time attestation must span the package, ensuring no rogue firmware is present on any die.
- **Distributed Initialization:** Training SerDes links, calibrating HBM interfaces, and initializing inter-chip protocols (e.g., UCle, CXL.cache/mem,

INTRODUCTION	1
BOOT-UP: ESTABLISHING THE INITIAL TRUST CHAIN	1
KEY CHALLENGES AT BOOT TIME	1
RUNTIME: OBSERVABILITY AND CONTROL IN A DISAGGREGATED WORLD	2
KEY RESPONSIBILITIES AT RUNTIME	2
THE ROLE OF TELEMETRY	2
FUTURE DIRECTIONS: AUTONOMOUS, PROGRAMMABLE MANAGEMENT FABRICS	3
EMERGING IDEAS	3
CONCLUSION	3
SECURITY REQUIREMENTS FOR SYSTEMS MANAGEMENT	3

proprietary fabrics) must occur in a cooperative sequence, often involving retry and fallback.

- **Out-of-Band Management:** The BMC's role expands to orchestrate chiplet coordination, but it may no longer reside on a monolithic baseboard. Virtual BMCs or embedded controllers within chiplets may emerge.

RUNTIME: OBSERVABILITY AND CONTROL IN A DISAGGREGATED WORLD

Once booted, the system enters a long-running operational phase where observability, telemetry, and adaptive control are critical for performance, reliability, and energy efficiency.

KEY RESPONSIBILITIES AT RUNTIME

- **Thermal and Power Management:** Power and heat must be managed not just at the socket or rack level, but within-chiplet. Thermal coupling between dies (e.g., logic and memory) complicates static budgeting and calls for real-time, closed-loop control.
- **Health Monitoring and Fault Containment:** Errors are no longer isolated to DIMMs or CPU cores. They may arise in interposers, chip-to-chip PHYs, or uncore components of a specific chiplet. Fault domains must be redefined dynamically.
- **Runtime Reconfiguration:** Workloads may require dynamic remapping of chiplet interconnects, on-the-fly changes in memory hierarchy, or even software-defined packet routing in AI fabrics. Management systems must support hot-plug semantics for dies.

THE ROLE OF TELEMETRY

Telemetry must shift from coarse-grained server-level data to fine-grained, real-time streaming from each chiplet interface. This includes:

- Slice-level bandwidth and latency counters
- Thermal maps across chiplet boundaries
- Error syndrome logs with location and temporal context
- Link quality metrics (e.g., BER, eye width) for inter-chiplet interconnects

These metrics must be queryable both reactively (e.g., during a failure) and proactively (e.g., via AI-based failure prediction).

FUTURE DIRECTIONS: AUTONOMOUS, PROGRAMMABLE MANAGEMENT FABRICS

To manage the growing complexity, we may need to treat systems management itself as a programmable domain—deploying microcontrollers or SmartNICs as autonomous agents within the fabric to enforce policies and execute recovery logic.

EMERGING IDEAS

- **Out-of-Band Observability Meshes:** Instead of relying on host CPUs for management, a lightweight observability network (potentially optical or RF-based) could collect telemetry from all chiplets with minimal latency.
- **Resilient, Self-Healing Fabrics:** Chiplet networks should tolerate failure by reconfiguring routing or reallocating work to healthy dies, ideally without reboot.
- **Epistemic Management Models:** In highly dynamic, multi-actor systems, a single source of truth may not exist. Management must shift to local knowledge models that reason probabilistically about the system state.

CONCLUSION

Managing systems from boot-up to runtime in chiplet-based datacenters requires a new mindset: one that treats the system as an ensemble of partially autonomous elements rather than a monolithic machine. From secure boot to dynamic fault recovery, every aspect of management must adapt to a more heterogeneous, interdependent, and fast-evolving hardware substrate.

The chiplet revolution is not just a hardware story—it is a software, systems, and observability revolution too.

SECURITY REQUIREMENTS FOR SYSTEMS MANAGEMENT

Security Protocol Top Level Requirements

Requirement	Comments
The operation of the security protocol shall be optional	Security comes with a cost. Security concerns vary across users and deployments. The option to pay or not pay the price of security is appropriate.
The security protocol shall allow for zero configuration startup	Zero configuration startup creates an IBE domain for a set of nodes. In the base case, all connected, discoverable nodes will be included. The IBE enrollment will enable every pair of participants to compute a pair-unique secret shared key.
The security protocol may employ node authentication with an external authentication service.	Where nodes are authenticated, the pluggable auth methods available over EAP, EAPoL and 802.1X are supported. A network with node authentication will not have zero configuration startup, because the node identifiers need to be enrolled into an authentication system.
The security shall provide data privacy for the operand data between nodes enrolled in an IBE group	The operator fields are plaintext so they do not get privacy. This is because they are processed by every link. Intermediate nodes do not have the keys necessary to decode the ciphertext.
The security protocol shall provide authentication that a node is the node enrolled during the common IBE enrollment phase.	
The security protocol shall provide frame integrity verification protecting against replay attacks and packet alteration.	

Authentication and Key Establishment Requirements

Requirement	Comments
The key establishment shall provide complete pairwise security between all enrolled nodes in an IBE group	The IBE system will enable each enrolled member to establish secret material from which a secret key for each node pair can be computed only by the members of that node pair.

Link Cipher Requirements

Requirement	Comments
The link cipher shall not increase frame sizes or reduce SDU sizes	It is common for link ciphers in IEEE 802 networks to consume bytes in each secured frame for nonces, keyID and ICVs. This either increases the size of the frame or reduces the number of bytes available for transporting data.
Reserved bits in SHF and STF shall be set to zero and checked to be zero on receipt.	An STF or SHF with set bits in a reserved field shall be treated as an integrity failure. Discussion in the meeting (Sept 10th 2025) was to ignore reserved bits. I prefer to make them part of the integrity check rules. This is open to change.
